WORKED SOLUTION

STEP II 2015, Question 4

Pure • Inverse Tangent Branches

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q4

The key idea

Knowing $\tan\theta$ only pins θ down up to a multiple of π, so each equation has infinitely many solution branches, all vertical translates of one another by π. Continuity and the value at $x = 0$ choose one branch, and continuity also forces us to *change* branch whenever the right-hand side runs off to $\pm\infty$: that is exactly the moment the angle passes through an odd multiple of $\frac{1}{2}\pi$. So in each part we first sketch the right-hand side, then follow the angle continuously.

Throughout, arctan means the principal value, with $-\frac{1}{2}\pi < \arctan u < \frac{1}{2}\pi$.

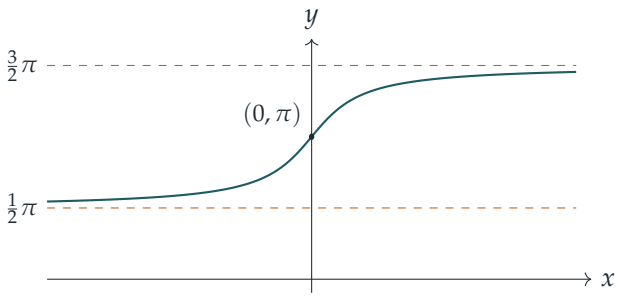
Solution

Part (i)

Every solution of $\tan\theta = x$ is $\theta = \arctan x + n\pi$ for an integer n. The function $\arctan x$ is continuous on the whole real line, so $\mathrm{f}(x) - \arctan x$ is a continuous function taking only values in $\{n\pi\}$; a continuous function cannot jump between these isolated values, so it is constant. Since $\mathrm{f}(0) = \pi$ and $\arctan 0 = 0$, the constant is π:

$$\mathrm{f}(x) = \pi + \arctan x.$$

So the curve is the usual arctan curve lifted by π: it increases strictly, passes through $(0, \pi)$ with gradient 1 (since $\mathrm{f}'(x) = \frac{1}{1+x^2}$), and has horizontal asymptotes $y = \frac{1}{2}\pi$ as $x \to -\infty$ and $y = \frac{3}{2}\pi$ as $x \to \infty$. It has rotational symmetry of order 2 about $(0, \pi)$, because $\mathrm{f}(-x) = 2\pi - \mathrm{f}(x)$.



Part (ii)

The curve $y = \dfrac{x}{1+x^2}$. Call this $q(x)$. It is odd, it is zero only at $x = 0$, and it has the sign of x. As $x \to \pm\infty$, $q(x) \approx 1/x \to 0$, so the x-axis is an asymptote, approached from above on the right and from below on the left. Differentiating,

$$q'(x) = \frac{(1+x^2) - 2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2},$$

so the stationary points are at $x = \pm 1$: a maximum $(1, \frac{1}{2})$ and a minimum $(-1, -\frac{1}{2})$. (Equivalently, $1 + x^2 \geq 2|x|$ by AM–GM, so $|q(x)| \leq \frac{1}{2}$.) The gradient at the origin is 1.

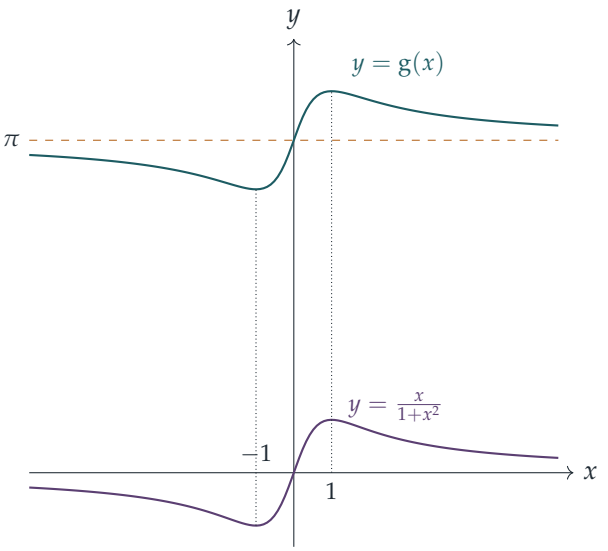
The curve $y = \mathrm{g}(x)$. Now the right-hand side $q(x)$ is continuous and bounded, so exactly the argument of part (i) gives

$$\mathrm{g}(x) = \pi + \arctan \frac{x}{1+x^2}.$$

Because arctan is increasing, g goes up and down exactly where q does. So

- $\mathrm{g}(x) \to \pi$ as $x \to \pm\infty$ (from above on the right, from below on the left);
- g has a maximum $\pi + \arctan \frac{1}{2} \approx 3.61$ at $x = 1$ and a minimum $\pi - \arctan \frac{1}{2} \approx 2.68$ at $x = -1$;
- it passes through $(0, \pi)$ with gradient $\dfrac{q'(0)}{1 + q(0)^2} = 1$.

Its shape is that of q, slightly flattened (since $\arctan u < u$ for $u > 0$) and lifted by π; it never comes near $\frac{1}{2}\pi$ or $\frac{3}{2}\pi$, so it never needs to change branch.



Part (iii)

The curve $y = \dfrac{x}{1-x^2}$**.** Call this $r(x)$. It is odd, it vanishes only at $x = 0$, and it has vertical asymptotes at $x = \pm 1$. As $x \to \pm\infty$, $r(x) \approx -1/x \to 0$. Its derivative is

$$r'(x) = \frac{(1-x^2)+2x^2}{(1-x^2)^2} = \frac{1+x^2}{(1-x^2)^2} > 0,$$

so r is strictly increasing on each of the three intervals where it is defined. Reading off the signs of x and $1-x^2$:

- on $x < -1$, r rises from 0^+ (as $x \to -\infty$) to $+\infty$ (as $x \to -1^-$);
- on $-1 < x < 1$, r rises from $-\infty$ to $+\infty$, through the origin with gradient 1;
- on $x > 1$, r rises from $-\infty$ (as $x \to 1^+$) to 0^- (as $x \to \infty$).

The curve $y = \mathrm{h}(x)$**.** On each interval separately, r is continuous, so $\mathrm{h}(x) - \arctan r(x)$ is a constant multiple of π there, but the multiple may be different on different intervals. We find it interval by interval.

On $-1 < x < 1$, the condition $\mathrm{h}(0) = \pi$ gives $\mathrm{h}(x) = \pi + \arctan r(x)$. As $x \to 1^-$, $r \to +\infty$, so $\mathrm{h}(x) \to \pi + \frac{1}{2}\pi = \frac{3}{2}\pi$; as $x \to -1^+$, $r \to -\infty$ and $\mathrm{h}(x) \to \frac{1}{2}\pi$. Continuity at $x = \pm 1$ forces $\mathrm{h}(1) = \frac{3}{2}\pi$ and $\mathrm{h}(-1) = \frac{1}{2}\pi$.

? **Ask yourself** Just to the right of $x = 1$ we have $r(x)$ large and negative, so $\arctan r(x)$ is close to $-\frac{1}{2}\pi$. Which multiple of π do we add to make h continuous at $x = 1$?

For $x > 1$ we have $\mathrm{h}(x) = n\pi + \arctan r(x)$, and as $x \to 1^+$ this tends to $n\pi - \frac{1}{2}\pi$. For this to equal $\frac{3}{2}\pi$ we need $n = 2$. Similarly, for $x < -1$ we have $\mathrm{h}(x) = m\pi +$

$\arctan r(x) \to m\pi + \frac{1}{2}\pi$ as $x \to -1^-$, and matching $\frac{1}{2}\pi$ needs $m = 0$. Hence

$$\mathrm{h}(x) = \begin{cases} \arctan \dfrac{x}{1-x^2}, & x < -1, \\ \pi + \arctan \dfrac{x}{1-x^2}, & -1 < x < 1, \\ 2\pi + \arctan \dfrac{x}{1-x^2}, & x > 1, \end{cases} \qquad \mathrm{h}(\pm 1) = \pi \pm \tfrac{1}{2}\pi.$$

So as r shoots up to $+\infty$ and reappears from $-\infty$, the angle h simply carries on increasing through $\frac{3}{2}\pi$ (and likewise through $\frac{1}{2}\pi$ at $x = -1$).

Now the features of the sketch.

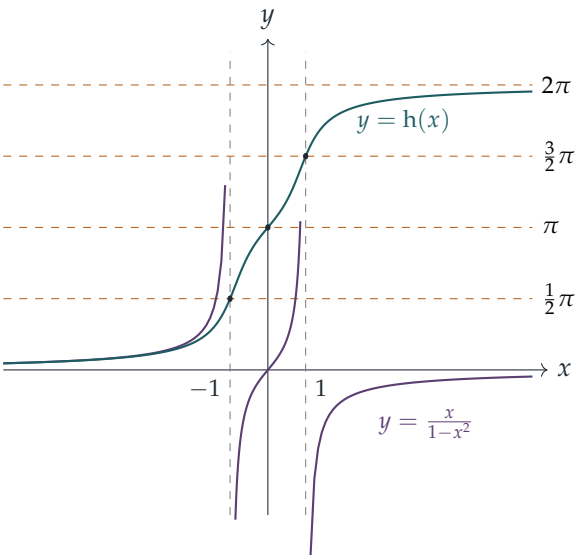
- *Asymptotes.* As $x \to -\infty$, $r \to 0^+$ so $\mathrm{h}(x) \to 0$ from above; as $x \to \infty$, $r \to 0^-$ so $\mathrm{h}(x) \to 2\pi$ from below.
- *Monotonic.* Away from ± 1, by the chain rule,

$$\mathrm{h}'(x) = \frac{r'(x)}{1 + r(x)^2} = \frac{1 + x^2}{(1-x^2)^2 + x^2} = \frac{1 + x^2}{x^4 - x^2 + 1} > 0.$$

  (The denominator is $(x^2 - \frac{1}{2})^2 + \frac{3}{4} > 0$.) This expression is perfectly well behaved at $x = \pm 1$, where it equals 2, so h is smooth there: the curve passes through $(1, \frac{3}{2}\pi)$ and $(-1, \frac{1}{2}\pi)$ with gradient 2, with no kink.
- *Key points.* $(0, \pi)$ with gradient 1; $(\pm 1, \pi \pm \frac{1}{2}\pi)$ with gradient 2.
- *Symmetry.* $\mathrm{h}(-x) = 2\pi - \mathrm{h}(x)$, so the curve has rotational symmetry about $(0, \pi)$.

So h increases steadily from 0 to 2π, an S-shaped curve like that of part (i) but covering twice the height. In the sketch the asymptotes $x = \pm 1$ of r line up exactly with the points where h crosses $\frac{1}{2}\pi$ and $\frac{3}{2}\pi$.

A tidy way to write all three cases at once: $\mathrm{h}(x) - \pi$ is the polar angle, in $(-\pi, \pi)$, of the point $(1 - x^2, x)$. That point never reaches the origin or the negative horizontal axis (when $1 - x^2 \leq 0$ we have $x \neq 0$), which is why the angle varies continuously.



Checks

- Each formula satisfies the defining equation: for instance $\tan(2\pi+\theta)=\tan\theta$, and the values at 0 are all π.
- Spot values: $g(1)=\pi+\arctan\frac{1}{2}\approx 3.605$, and $h(2)=2\pi+\arctan(-\frac{2}{3})\approx 5.695$, which lies between $\frac{3}{2}\pi\approx 4.712$ and $2\pi\approx 6.283$ as the sketch requires.
- Continuity of h: $h(0.999)\approx 4.710$ and $h(1.001)\approx 4.714$, either side of $\frac{3}{2}\pi\approx 4.712$, consistent with gradient 2 at $x=1$. A computer check stepping through $[-10,10]$ confirmed that none of f, g, h jumps anywhere.

Common slips

- Writing $h(x)=\pi+\arctan\frac{x}{1-x^2}$ for all x: that jumps down by π at $x=1$ and up by π at $x=-1$, so it is not continuous.
- Drawing a vertical asymptote in $y=h(x)$ at $x=\pm1$ just because $\frac{x}{1-x^2}$ has one; the angle stays finite, only its tangent blows up.
- Showing a corner on $y=h(x)$ at $x=\pm1$; the gradient there is 2 from both sides.

Beyond STEP

What we have done is choose a continuous *branch* of a many-valued inverse function, and the rule "add π each time the tangent passes through infinity" is a first taste of lifting a path to a covering space. The same idea, for the complex logarithm and $\arg z$, underlies the winding number and the argument principle in complex analysis.

Needham's *Visual Complex Analysis* develops this geometrically and is a delightful read.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.